

SportsLink: Technical Specification

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1. Purpose, scope and phases

This spec tells the tech team what to build for SportsLink: one mobile app (iOS and Android) and one web app for players and vendors, plus a separate web admin panel. Business rules come from the SportsLink Company Foundation Document. Where a decision is still open, it is marked **OPEN** and the team should build it as a configurable setting, not hard-code it.

Build principles

- One account, two modes: Player and Vendor. Admin is a separate app with separate login.
- Multi-country ready: currency, timezone, language, payment methods and commission rules stored per country.
- Every business rule that may change (commission, hold times, billing deadlines, radius limits) is an admin setting.
- No card data or player money passes through SportsLink servers in the current model.

Phases

Phase	Scope
1. Launch	Auth and verification, profiles, vendor onboarding, venues and courts, booking with advance and split bills, manual bookings, create and join match, chat, vendor billing, core admin panel
2. Growth	Find Players live, skill rating (Glicko-2), behaviour reviews, teams, leaderboards, vendor analytics, subscriptions, ads, promo codes
3. Tournaments	Tournament management, brackets, results, tournament rating, detailed stats
4. Future	Government trials, marketplace, add-ons, new countries

The data model in section 4 covers all phases so later phases do not need a rebuild.

2. Recommended tech stack

One language (TypeScript) across mobile, web, admin and backend, so a small team can move fast and share code. Versions should be the current stable ones when work starts.

Layer	Recommendation	Why
Mobile app	React Native with Expo	One codebase for iOS and Android, over-the-air updates
Web app (player and vendor)	Next.js (React)	Shares components and logic with mobile, good SEO for venue pages
Admin panel	Next.js, separate deployment	Isolated from the public app for security
Backend API	Node.js with NestJS	Structured modules, strong typing, easy to test
Database	PostgreSQL with PostGIS	Reliable transactions for bookings, fast distance queries for Find Players
Cache, locks, queues	Redis with BullMQ	Slot holds, rate limits, background jobs
Realtime	Socket.IO on the backend, Redis adapter	Chat, live Find Players, live booking updates
Search	PostgreSQL full text at launch, OpenSearch later if needed	Keep launch simple
File storage	S3-compatible storage with signed URLs	CNIC images private, venue photos public via CDN
Push notifications	Firebase Cloud Messaging (Android and iOS via APNs)	Standard and free
SMS OTP	A Pakistan SMS provider with a fallback provider	Local delivery rates are better. OPEN: provider choice
Maps	Google Maps Platform	Best coverage in Pakistan. Watch cost limits
Hosting	AWS, nearest region to Pakistan (Bahrain or Mumbai)	OPEN: confirm data residency rules with the lawyer

Layer	Recommendation	Why
Monitoring	Sentry for errors, CloudWatch or Grafana for metrics	Catch problems before users report them
CI/CD	GitHub Actions, Expo EAS for app builds	Automated tests and releases

Monorepo (for example Turborepo) with shared packages for types, validation schemas, API client and UI components.

3. System architecture

A modular monolith at launch: one backend with clear internal modules, split into services only when load requires it.

```

flowchart LR
    M["M[Mobile app<br/>React Native]"] --> G["G[API gateway<br/>REST + WebSocket]"]
    W["W[Web app<br/>Next.js]"] --> G
    A["A[Admin panel<br/>Next.js]"] --> G
    G --> B["B[Backend<br/>NestJS modules]"]
    B --> P["P[(PostgreSQL<br/>+ PostGIS)]"]
    B --> R["R[(Redis<br/>locks, cache, queues)]"]
    B --> S["S[(File storage<br/>+ CDN)]"]
    B --> J["J[Workers<br/>billing, ratings, alerts]"]
    J --> N["N[Push + SMS<br/>FCM, OTP provider]"]
  
```

Backend modules: auth, users, verification, venues, bookings, payments, billing, matches, find-players, chat, ratings, teams, tournaments, notifications, moderation, ads, subscriptions, admin, audit.

4. Core data model

Main tables and their key fields. All tables have id (UUID), created_at, updated_at, and soft delete where records must be kept for audit. Money is stored as integer minor units with a currency code.

Entity	Key fields	Notes
country	code, currency, timezone, languages, enabled_payment_methods	Expansion ready

Entity	Key fields	Notes
user	phone, name, dob, gender, city, photo, status, is_minor, guardian_user_id	One user can be player and vendor
verification	user_id, doc_type (CNIC, B-Form), front_url, back_url, status, reviewer_id	Images private, encrypted
player_sport	user_id, sport_id, position, self_level, available_now, alert_mode	alert_mode: always or only_when_available
sport	name, team_size_min, team_size_max, stat_schema	stat_schema defines detailed stats per sport
vendor	owner_user_id, business_name, billing_model, commission_rate, monthly_fee, credit_balance, status	billing_model: percentage or monthly
vendor_staff	vendor_id, user_id, branch_ids, permissions	Limited access
branch	vendor_id, name, address, location (geography point), status	Multiple per vendor
court	branch_id, sport_ids, name, surface, photos, slot_minutes, status	A court or ground
price_rule	court_id, day_type, start_time, end_time, price	Peak, off-peak, weekend, holiday
venue_policy	branch_id, advance_type, advance_value, cancel_refund, noshow_refund, cancel_window_hours, recurring_allowed	Vendor controlled
payment_account	vendor_id, method, account_title, account_number, status, approved_by	Admin approval required
site_visit	branch_id, requested_by, scheduled_at, visitor_admin_id, result, notes, photos	Required for approval

Entity	Key fields	Notes
booking	court_id, start_at, end_at, source (app, manual), status, total, advance_due, created_by, match_id, recurring_series_id, counts_for_billing	See section 6
booking_share	booking_id, user_id, amount, status, txn_reference, confirmed_by	Per-player bill
recurring_series	court_id, weekday, start_time, weeks, status	All weeks paid upfront
match	host_id, sport_id, booking_id or unlisted_venue, start_at, slots_total, slots_filled, filters, visibility, status	See section 8
match_player	match_id, user_id, team_side, status, paid_by	status: requested, approved, confirmed, removed, left
player_request	user_id, sport_id, location, radius_km, players_needed, window, filters, status, expires_at	Find Players
conversation, message	type, members, body, media, flagged_phone	Chat
result	match_id, submitted_by, scores, status, confirmed_by, dispute_id	Both sides confirm
rating	user_id or team_id, sport_id, kind (skill, tournament), value, deviation, volatility, games	Glicko-2 fields
review	match_id, from_user, to_user, stars, tags, comment	Behaviour only
team, team_member	sport_id, captain_id, members, role	Team rating
tournament, fixture, entry	format, sport, fee, prize, status, bracket, results	Admin only
invoice, invoice_line	vendor_id, period, lines, amount, status, due_at, paid_at	Monthly billing
credit_txn	vendor_id, type (topup, deduction, adjustment), amount, booking_id	If prepaid credit is chosen

Entity	Key fields	Notes
report, moderation_action	target, reason, status, action, admin_id	Trust and safety
audit_log	actor_id, role, action, target, before, after, ip	Append only
setting	key, value, country_code	All admin configurable rules

5. Auth, verification and accounts

Sign-up: phone number, OTP, then profile. Sessions use short-lived access tokens and rotating refresh tokens, stored in secure device storage.

Verification: user uploads CNIC front and back (minors: B-Form, plus guardian link). Status goes pending, approved or rejected with a reason. **OPEN:** whether CNIC is required at sign-up or only before creating, joining or using Find Players. Build it as a setting.

Minors: date of birth decides minor status. A minor account stays locked until a guardian with an approved CNIC accepts consent in their own account. Consent text version and timestamp are stored. Minor safeguards from the Foundation Document section 10.2 are feature flags, pending your decision.

Mode switch: a user becomes a vendor by submitting a vendor application. The app shows a Player or Vendor switch once approved. Staff accounts only see the vendor branches and actions granted to them.

Edge cases

- OTP: 60 second resend timer, 5 attempts then 30 minute lockout, rate limit per phone and per device.
- Phone number change: OTP on both old and new numbers, or admin review if old number is lost.
- Same CNIC on two accounts: block the second and flag for moderation.
- Banned users: device and CNIC are recorded so a new phone number cannot create a fresh account.
- Minor turns 18: guardian link ends automatically, user is asked to verify with CNIC.
- Account deletion: personal data removed, bookings, invoices and audit records kept anonymised.

6. Booking engine

Double booking must be impossible. App bookings, manual bookings, maintenance blocks and recurring series all live in the same booking table, protected by a database exclusion constraint on court and time range, plus a Redis lock during checkout.

6.1 Booking states

```
stateDiagram-v2
    [*] --> held: slot selected
    held --> pending_payment: shares created
    held --> expired: hold timeout
    pending_payment --> confirmed: vendor confirms advance
    pending_payment --> expired: payment timeout
    confirmed --> completed: slot ends
    confirmed --> cancelled: player or vendor cancels
    confirmed --> no_show: vendor marks
    completed --> [*]
```

Manual bookings by the vendor go straight to confirmed.

6.2 Rules

- Hold time and payment confirmation time are settings (suggested 15 and 60 minutes).
- Slot price is calculated from price rules at booking time and stored on the booking, so later price changes do not affect it.
- Advance is set by venue policy: fixed amount or percentage.
- Recurring booking: only if the venue allows it. All weeks are created and checked together. If any week clashes, the player sees which ones and can skip them or cancel.
- Times are stored in UTC and shown in the venue's timezone.

6.3 Edge cases

- Two players select the same slot at once: the lock gives it to the first, the second sees it taken immediately.
- Vendor blocks a court for maintenance over existing bookings: not allowed until those bookings are moved or cancelled with notice.
- Vendor changes opening hours: existing bookings stay, new slots follow new hours.
- Venue suspended or banned: future bookings are flagged, players notified, refunds follow admin instruction.
- Vendor cancels a confirmed booking: player gets a full refund regardless of policy (recommended rule, OPEN), and the cancellation counts against the venue's reliability.

- Player cancels: refund follows the policy shown at booking time, even if the vendor changes it later.
- Vendor never confirms payment: booking expires, player is notified, support ticket opened automatically if a transaction ID was entered.
- Rain or closure: vendor can mark a slot cancelled by venue, which triggers the vendor cancel rule above.
- Player wants to extend: only if the next slot on the same court is free, charged as a new linked booking.

7. Payments and vendor billing

Players pay the vendor directly. SportsLink stores payment references and confirmations only, never card or wallet credentials.

7.1 Payment flow

```
sequenceDiagram
    participant P as Player
    participant A as SportsLink
    participant V as Vendor
    P->>A: Select slot, create share
    A-->>P: Show vendor's approved account
    P->>V: Pay advance (JazzCash, Easypaisa, bank)
    P->>A: Submit transaction ID
    A->>V: Ask to confirm receipt
    V->>A: Confirm or reject
    A-->>P: Booking confirmed
```

- Cash at venue: booking confirmed only if the venue allows unpaid bookings. Otherwise an advance is required.
- Duplicate transaction IDs across bookings are rejected and flagged.
- A vendor rejecting a submitted payment opens a dispute automatically.

7.2 Split bills

- A booking has one share per paying player. The host's share covers the players they brought.
- Joining players get their own share when approved.
- Booking is confirmed when the confirmed shares cover the advance.
- Unpaid share after its deadline: the host is notified and can remove and replace that player.

7.3 Refunds

- Refund amount calculated from the policy stored on the booking.
- Vendor sends the refund directly, marks it sent with a reference. Player confirms received or raises a dispute.

7.4 Vendor billing

- `counts_for_billing` is true for app bookings and for manual bookings (rate for manual bookings **OPEN**).
- Percentage model: $\text{commission} = \text{rate} \times \text{booking total}$, on completed bookings (and no-shows, **OPEN**).
- Monthly plan: fixed fee per vendor, set by admin.
- Invoice job runs on the 1st of each month in the venue's timezone. Invoice is immutable once issued; corrections are credit notes.
- Overdue ladder, all settings: reminder at 7 days, warning at 14, hidden from search at 21, blocked at 30.
- Vendor pays by bank transfer or wallet to SportsLink and uploads proof. Finance marks it paid.
- **If prepaid credit is chosen (OPEN):** commission is deducted from `credit_balance` when a booking is confirmed. At zero, new app bookings pause. Refunded bookings return the commission to credit.

7.5 Subscriptions

- Player ad-free subscription through Apple and Google in-app purchases on mobile, and a web payment provider on web. Entitlements are stored server side and synced from store notifications.

8. Matches

A match links a host, a sport, a time and either a booking at a listed venue or an unlisted venue address. The host approves every join request.

```
stateDiagram-v2
    [*] --> open: host creates
    open --> full: all slots confirmed
    full --> open: a player leaves
    open --> cancelled: host cancels
    full --> in_progress: start time
    open --> in_progress: start time
    in_progress --> result_pending: end time
    result_pending --> completed: both sides confirm
    result_pending --> disputed: sides disagree
    disputed --> completed: admin decides
```

8.1 Rules

- Host sets slots_total and players already confirmed. Joiners fill the rest.
- Filters: skill rating range, age range, gender, verified only. Players outside the filters cannot see or request the match.
- Join request states: requested, approved, confirmed (share paid), declined, withdrawn, removed.
- Approval deadline: requests not answered before a cut-off (setting, for example 2 hours before start) expire.
- Unlisted venue: show the warning screen every time before creating or joining. No booking or payment is created.

8.2 Edge cases

- Host cancels: every confirmed player is notified. Refunds follow the booking policy. Frequent cancelling lowers host reliability.
- Host leaves but others want to play: host can transfer the host role to a confirmed player.
- Player leaves after paying: refund follows policy. Slot reopens and the host is notified.
- More approvals than slots: the system blocks approval once slots are full, remaining requests go to a waitlist.
- Player joins two matches at overlapping times: warn and block.
- Player blocked by the host: cannot see that host's future matches.
- Nobody joins: host is reminded before the payment deadline and can cancel without penalty (window is a setting).
- Match at an unlisted venue with a reported safety issue: moderation can hide all matches at that address.

9. Find Players (live)

A player broadcasts a request, nearby matching players get a push notification, they accept, and the requester picks who joins. Everything after that moves to a chat.

9.1 Flow

1. Requester sets sport, players needed, radius, time window (now, today, custom) and filters.
2. Backend queries players with that sport whose last known location is inside the radius (PostGIS `ST_DWithin`), who match the filters, and whose alert mode allows it.
3. Notifications go out in batches, nearest first, so the requester is not flooded.
4. Accepting players appear to the requester live over WebSocket with rating, reliability and approximate distance.
5. Requester selects players. A group chat opens. Unselected players are told politely.
6. In chat they agree venue and time, then convert the request into a match (listed venue booking or unlisted venue).

9.2 Location

- Location is sent only while the app is open, or in the background if the user turns on Available to play. Precise location is never stored long term, only a rounded location (about 500 m) for matching.
- Other users only see distance bands (for example under 2 km, 2 to 5 km), never coordinates.

9.3 Limits and edge cases

- Radius: user chosen, capped by a setting (for example 1 to 25 km).
- Rate limits: requests per user per hour, and alerts received per user per day (settings).
- Quiet hours: no alerts at night unless the user allows it.
- Request expires at the end of its window, or when the requester closes it.
- Requester cancels after selecting players: chat stays open briefly, players notified.
- Selected player goes silent: requester can remove them and pick another from the accepted list.
- Nobody nearby: suggest widening the radius, or show open matches nearby.
- Minor safeguards apply here (Foundation Document 10.2, feature flagged).

10. Chat

Chat supports one-to-one and group conversations tied to matches, teams, Find Players requests and bookings (player to vendor).

- Message types: text, photo, voice note, location pin, system message (for example "Ali joined the match").
- Delivery over WebSocket with push fallback. Read receipts and typing indicator.
- Messages stored server side so moderators can review reported conversations. Not end-to-end encrypted, which is needed for moderation. State this in the privacy policy.
- **Phone number detection:** detect Pakistani number patterns (for example 03xx xxxxxxx, +92), including spaced or written-out digits. Show a safety warning, then send only if the user confirms. Log the event.
- Media: size limits, image compression, malware scanning on upload.
- Block: blocked users cannot message or see each other's activity.
- Report: reporter picks a reason, the last messages are attached to the report automatically.
- Retention: group chats stay readable for a set period after the match (setting), then archive.
- Minor accounts: private chat rules follow the minor safeguards feature flags.

11. Ratings, results and disputes

Skill ratings use Glicko-2, one rating per user per sport, and one per team per sport.

Tournament ratings are a separate `kind` updated only from admin-entered tournament results.

11.1 Result flow

1. After the match ends, any confirmed participant submits the result (winner, draw, scores, optional detailed stats).
2. The other side confirms or disputes within 48 hours (setting). Silence after the deadline counts as confirmed, **OPEN**.
3. Dispute: both sides add a note, admin decides, decision is final and logged.
4. Only completed matches with check-in or both-side confirmation update ratings.

11.2 Glicko-2 settings

- Start rating 1500, rating deviation 350, volatility 0.06, system constant tau 0.5. All configurable.

- Rating period: process each result as it completes (a period of one game), which is common for apps.
- Deviation grows slowly when a player is inactive, so returning players' ratings adjust faster.
- Library: use a well-tested open source Glicko-2 implementation, wrapped with unit tests from the published example in Glickman's paper.

11.3 Team sports

- Each side gets a composite rating: mean of member ratings, deviation combined from member deviations.
- Each player is updated as if they played one game against the other side's composite.
- Team entities (section 12) get their own rating updated the normal way.

11.4 Anti-abuse

- Repeat opponents: after N games against the same people in a period, rating change is reduced (setting).
- Rating changes are held until the confirmation window closes.
- Admin can void a match and recalculate affected ratings.

11.5 Tiers and leaderboards

- Tiers mapped from rating bands (for example Bronze, Silver, Gold, Platinum, Elite). Bands are a setting.
- Players with high deviation show as "Provisional" and are hidden from leaderboards until they settle.
- Leaderboards per sport, computed by a scheduled job and cached.

11.6 Behaviour reviews

- Allowed only between confirmed participants, once per pair per match, within 48 hours.
- Stars (1 to 5) plus tags. Shown as an average and top badges. Never affect skill rating.

12. Teams and tournaments

12.1 Teams

- Created by a player for one sport. Captain invites members, members accept.
- Roles: captain, vice captain, member. Captain can transfer the role.

- Team page: roster, team rating, match history, tournament history.
- Team matches: a team creates a match against another team, or opens it for any team to challenge.
- Edge cases: member leaves mid-tournament (roster locked after registration closes, admin can approve changes), captain banned (vice captain takes over, else admin assigns).

12.2 Tournaments (admin only)

- Admin sets sport, format, individual or team entry, entry fee, prize, venue, dates, eligibility (rating range, age, gender), max entries and registration deadline.
- Formats: knockout, league, round robin, groups then knockout. Bracket generation with seeding by tournament rating, random for unrated.
- Registration: player or captain registers and pays the fee. **OPEN**: fee paid to SportsLink or to the host venue, given the no-holding-money model.
- Fixtures, live standings and results entered by admin. Walkovers and withdrawals handled by admin.
- Tournament rating updated from official results only.
- Government trials: a placeholder screen marked Coming soon. Data model keeps a `programme` field on tournaments for future trials.

13. Vendor module

Available in the same mobile app (Vendor mode) and on web, where the calendar and analytics are easier to use.

13.1 Onboarding

```

flowchart LR
    A[Submit CNIC, bank,<br/>venue details] --> B[Ops review]
    B --> C[Site visit<br/>requested or scheduled]
    C --> D{Visit result}
    D -->|Pass| E[Approved, live]
    D -->|Fail| F[Rejected with reason]
    B -->|Missing info| A
  
```

Bank account title must match the CNIC name. Ownership or lease document is optional. Any later change to payment accounts goes back to admin approval, and the old account stays active until the new one is approved.

13.2 Features

- Branch, court and price rule management, photo gallery, facilities, rules.
- Policies: advance, cancellation refund, no-show refund, cancel window, recurring allowed.
- Calendar per branch and court, day and week views, drag to block time.
- Manual booking with customer name and optional phone (stored privately, never shown to players).
- Payment confirmation queue, no-show marking, refund recording.
- Staff management with permissions: view bookings, create bookings, confirm payments, edit prices, view revenue, manage staff. Scoped by branch.
- Analytics: bookings, occupancy heatmap by hour, revenue by court and source, cancellations, rating trend, reviews with reply.
- Revenue calculator: projected revenue from price rules and occupancy.
- Billing: current month running total, invoices, payment upload, credit balance if prepaid is chosen.

13.3 Edge cases

- Vendor deletes a court with future bookings: blocked until those bookings are resolved.
- Vendor price change: applies only to new bookings.
- Staff removed: sessions revoked immediately.
- Vendor blocked for non-payment: listings hidden, existing confirmed bookings still honoured and visible to the vendor.

14. Admin panel and permissions

Separate web app, separate domain, admin login with password plus mandatory two-factor authentication. Optional IP allow-list.

Permission area	Owner	Super admin	Operations	Finance	Moderation
Manage admins and roles	Yes	Yes, except Owner	No	No	No
Venue approval, site visits	Yes	Yes	Yes	No	No
Payment account approval	Yes	Yes	Yes	Yes	No
Commission, plans, invoices	Yes	Yes	View	Yes	No
Player verification	Yes	Yes	Yes	No	Yes
Bans and suspensions	Yes	Yes	Venues only	No	Yes

Permission area	Owner	Super admin	Operations	Finance	Moderation
Reported chats and reviews	Yes	Yes	No	No	Yes
Disputes (results, bookings)	Yes	Yes	Yes	Refund disputes	Yes
Tournaments	Yes	Yes	Yes	Fees view	No
Promo codes, campaigns, ads	Yes	Yes	Yes	View	No
Analytics	Yes	Yes	Operational	Financial	Safety
Settings and feature flags	Yes	Yes	No	Billing settings	No

Permissions are stored as granular actions, so roles can be adjusted later without code changes.

Requirements

- Every admin action writes to the append-only audit log with before and after values.
- Reading a private chat or CNIC image is itself logged.
- Bans need a reason and optional expiry. Ban types: warning, temporary suspension, permanent ban, shadow limit on Find Players.
- Auto flags into queues: venue rating below a threshold, many no-shows, many reports, duplicate CNIC, payment account changes.
- Dashboards: bookings, commission due and collected, outstanding invoices, active users, new venues, top sports, city breakdown.

15. Notifications

All notifications go through one notification service with templates per language, user preferences, quiet hours and delivery logs.

Event	Player	Vendor	Channel
Booking held, confirmed, cancelled, reminder	Yes	Yes	Push, in-app
Payment submitted, confirmed, rejected	Yes	Yes	Push, in-app
Match join request, approved, declined, removed	Yes	No	Push, in-app
Find Players nearby request	Yes	No	Push
New chat message	Yes	Yes	Push, in-app
Result to confirm, dispute update	Yes	No	Push, in-app

Event	Player	Vendor	Channel
Review received	Yes	Yes	In-app
Invoice issued, reminder, overdue, blocked	No	Yes	Push, in-app, SMS for final warning
Verification approved or rejected	Yes	Yes	Push, SMS
Minor activity (guardian)	Guardian	No	Push
Tournament updates	Yes	No	Push, in-app
Marketing campaigns	Opt-in only	Opt-in only	Push

16. Security and privacy

- **Secrets:** never in code or the repo. Use a secrets manager (for example AWS Secrets Manager). Secret scanning in CI.
- **Encryption:** TLS everywhere. Database and file storage encrypted at rest. CNIC images and numbers encrypted at field level with separate keys.
- **Access to sensitive data:** CNIC images served only through short-lived signed URLs to admins with permission. Every view is logged.
- **Phone numbers:** never returned by any player-facing API.
- **Location:** rounded before storage, never exposed as coordinates to other users.
- **API protection:** rate limiting per IP, user and device. Input validation with shared schemas. Protection against the OWASP Top 10.
- **Authorisation:** every endpoint checks ownership or role. Vendor staff scoped to their branches. Tests for access rules on every endpoint.
- **Fraud signals:** duplicate CNIC, duplicate device, duplicate transaction IDs, unusual rating gains, sudden payment account changes.
- **Backups:** daily automated backups, point in time recovery, restore tested every quarter.
- **Privacy:** privacy policy covering CNIC, location, chat storage and ads. Data export and deletion on request.
- **Penetration test** before public launch.

17. Non-functional requirements

Targets for launch. They should be revisited once real usage data exists.

Area	Target
API response time	95% of requests under 300 ms
Slot availability accuracy	Zero double bookings, verified by concurrency tests
Find Players alert delivery	First alerts sent within 5 seconds of a request
Chat delivery	Under 1 second when both users are online
Uptime	99.5% monthly at launch
App performance	Cold start under 3 seconds on a mid-range Android phone
Network	Usable on slow 3G, with retries and offline states
Devices	Android 8 and later, iOS 15 and later, latest two versions of major browsers (confirm when starting)
Accessibility	Readable text sizes, sufficient contrast, screen reader labels
Localisation	English at launch, structure ready for Urdu with right-to-left layout (OPEN)
Scalability	Launch city load, designed to scale horizontally for national rollout

18. Delivery

Phase 1 is about 10 to 12 weeks with the team below, including 2 weeks of design and setup. A 6-week target is only possible for a cut-down pilot (Foundation Document section 12). Estimates are approximate.

18.1 Suggested team

Role	Count	Focus
Tech lead / backend	1	Architecture, booking engine, billing
Backend developer	1	Matches, chat, notifications, admin APIs

Role	Count	Focus
Mobile developer (React Native)	1 to 2	Player and vendor app
Web developer (Next.js)	1	Web app and admin panel
UI/UX designer	1 (part time after week 3)	Flows, designs, design system
QA engineer	1	Test plans, concurrency and payment flow testing

18.2 Environments

- Local, development, staging (production-like data volumes, fake data only), production.
- No real CNIC images, phone numbers or payment details outside production.

18.3 Testing

- Unit tests for pricing, commission, billing, Glicko-2 and permissions.
- Integration tests for booking, split payment and match flows.
- Concurrency tests for double-booking prevention.
- End-to-end tests for main journeys on Android, iOS and web.
- Linting and type checks block merges. Test failures are reported, never skipped.

18.4 Phase 1 done means

- ☐ A player can sign up, verify, find a venue, book with split payment and get confirmation.
- ☐ A vendor can apply, pass a site visit, list courts with prices, and see app and manual bookings in one calendar.
- ☐ A host can create a match, approve players, and each player pays their share.
- ☐ Players can chat, with the phone number warning working.
- ☐ Admin can approve venues and payment accounts, ban users, set commission per vendor and issue monthly invoices.
- ☐ No double booking under concurrency testing.
- ☐ Security checklist and penetration test passed.

19. Open technical questions

Build each as a setting or feature flag until decided.

- ☐ Billing model: postpaid invoices, prepaid vendor credit, or collecting the advance through a licensed gateway?
- ☐ Commission rate on manual bookings and on no-shows?
- ☐ Does silence after the result window count as confirmation?
- ☐ Vendor cancellation: always a full refund to the player?
- ☐ CNIC required at sign-up or only before creating, joining or Find Players?
- ☐ Which minor safeguards are switched on?
- ☐ Tournament fees: paid to SportsLink or to the venue?
- ☐ SMS OTP provider and backup provider?
- ☐ Hosting region, once data residency is confirmed by the lawyer?
- ☐ Urdu at launch or later?
- ☐ Launch sports list and launch city?
- ☐ Ad network or direct sponsor ads only?